Paper 73 — ADOPT

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Modified Adam can converge with any β_2 with the optimal rate

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The paper in one minute

Question: Why can Adam fail to converge in theory for fixed choices of its second-moment decay β_2?
Model/Method: Reorder moment estimation so the current stochastic gradient is normalized with the *previous* second-moment estimate, and apply momentum after normalization.
Finding: ADOPT attains an $O(T^{-1/2})$ first-order convergence measure for any allowed β_2, under bounded second moments and smoothness, without assuming almost-sure bounded stochastic gradients.
Why we care: A subtle dependence between current noise and adaptive denominators can matter as much as the optimization landscape itself.

1. Understand the dependency problem

Adam uses first-moment momentum m_t and a second-moment scale v_t based on squared recent gradients. Roughly, it moves like $-m_t/\sqrt{v_t}$. But v_t already contains the *same* random sample g_t whose effect it rescales. Thus the random direction and random denominator are coupled, complicating a guaranteed descent argument. Think of estimating a student's exam score and then using that same score to choose how much to scale it. The scaling is now statistically dependent on the observation. The algebraic issue here is conditional dependence, not that the use of g_t in a denominator is always intrinsically bad.

2. Worked example: the ordering really changes the step

Suppose the previous second-moment estimate is $v_{t-1} = 1$, the new scalar gradient is $g_t = 4$, and $\beta_2 = 0.9$ (ignore ϵ and momentum for illustration). A same-step denominator gives

$$v_t = 0.9(1) + 0.1(4^2) = 2.5, \qquad \frac{g_t}{\sqrt{v_t}} = \frac{4}{\sqrt{2.5}} \approx 2.53.$$

ADOPT instead divides the *current* gradient by the *previous* denominator:

$$\frac{g_t}{\sqrt{v_{t-1}}} = \frac{4}{1} = 4.$$

The point is *not* that the ADOPT step should always be bigger. It is that v_{t-1} is measurable using past samples, so it does not depend on the fresh noise g_t conditional on the history.

Key first-principles connection
If g_t is unbiased conditionally on past information, then multiplying it by a past-computed scale preserves an analyzable conditional expectation. A denominator that already uses g_t breaks this simple separation.

3. Why simply lagging the denominator is not enough

For RMSProp without momentum, the lagged denominator already eliminates the most direct current-sample coupling. With Adam-style momentum, however, the momentum vector itself is an accumulation of earlier gradients that are also embedded in the adaptive scale. The authors therefore change the *order* of the operations:

$$\boxed{m_t = \beta_1 m_{t-1} + (1-\beta_1)\frac{g_t}{\max\{\sqrt{v_{t-1}}, \epsilon\}}},$$

$$\theta_t = \theta_{t-1} - \alpha_t m_t, \qquad v_t = \beta_2 v_{t-1} + (1-\beta_2) g_t^2.$$

The componentwise formula above captures the important dependency structure; Algorithm 1 specifies initialization and the precise ordering used in the implementation. The fresh gradient is normalized using history, and its normalized version is accumulated into momentum. By comparison, Adam forms momentum first and then normalizes it.

4. What is the actual mathematical guarantee?

Theorem 4.1 assumes a lower-bounded objective, unbiased stochastic gradients, L-smoothness, and a finite *second moment* $\mathbb{E}\|g_t\|^2 \le G^2$ (Assumptions 2.1–2.3, 2.5). With $\alpha = \Theta(T^{-1/2})$ it controls a first-order stationarity measure:

$$\min_{1\le t\le T} \left(\mathbb{E}\|\nabla f(\theta_{t-1})\|^{4/3}\right)^{3/2} = O(T^{-1/2}).$$

The exponent and expectation are part of the theorem, so it should not silently be rewritten as a simple bound on $\mathbb{E}\|\nabla f\|^2$. Importantly, the authors remove the stronger almost-sure assumption $\|g_t\| \le G$; they *do not* remove all noise or moment assumptions.

Why this belongs in the survey
It addresses convergence of stochastic adaptive updates, not second-order stationarity. The paper does not prove that ADOPT escapes strict saddles in finite time or finds global minima.

5. Evidence, contributions, and limitations

Contributions. (i) Diagnose the problematic dependence between the current stochastic gradient and adaptive scaling; (ii) design ADOPT's lagged normalization-before-momentum update; (iii) establish a fixed-β_2 $T^{-1/2}$ convergence result under finite second-moment conditions; (iv) compare against Adam-type baselines across image classification, generative, language, and reinforcement-learning tasks. **Experimental reading.** The paper reports improvements on several benchmarks, but the exact gains depend on task and hyperparameters. When citing these results, distinguish "reported better performance" from the mathematical convergence theorem.

6. Important assumptions and cautions

- The stochastic gradient must be unbiased, with bounded second moment; this is less restrictive than almost-sure bounded gradients but is still a condition.
- Smoothness and a lower-bounded objective support the descent analysis; the rate concerns a first-order measure.
- "Any β_2" is not a promise about arbitrary learning rates, optimizer settings, or batch sequences.
- ADOPT does not provide a general saddle-escape theorem, generalization guarantee, or superiority over Adam on every model.

7. Quick reference

Adam coupling	Current g_t contributes to both numerator and current v_t.
ADOPT normalization	Normalize g_t with v_{t-1} first; then update momentum.
Moment assumption	$\mathbb{E}\|g_t\|^2 \le G^2$ (not almost-sure bounded g_t).
Rate	Theorem 4.1: $O(T^{-1/2})$ for the specified gradient-norm measure.
Guarantee class	Approximate first-order stationarity, not saddle escape.

Survey bridge / Papers Master note

ADOPT modifies Adam by normalizing the fresh gradient with the previous second moment before applying momentum. It proves an $O(T^{-1/2})$ first-order stationarity measure for any fixed β_2 under smoothness and finite-second-moment assumptions. Useful for adaptive stochastic dynamics, not a direct second-order saddle-escape result.